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# Well-solved cases of the 2-peripatetic salesman problem

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# WELL-SOLVED CASES OF THE 2-PERIPATETIC SALESMAN PROBLEM

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The 2-peripatetic Salesman Problem (2-PSP for short) is the problem, where 2 edge-disjoint Hamiltonian cycles of minimal total length are required. We have investigated whether the 2-PSP, either the sum version or the bottleneck version can also be well-solved in known well-solved cases of the TSP.

We show how to polynomially solve the 2-PSP in the following non-trivial cases:

- problems on Small matrices with distinct values for the sum as well as for the bottleneck criterion;
- problems on asymmetric Circulant matrices with a prime number of cities again for both criteria;
- problems with the bottleneck criterion on max-distribution matrices with the number of negative $a$-values equal to the number of nonnegative $b$-values.

*Keywords:* Traveling salesman problem; peripatetic salesman problem; edge disjointness; well-solved cases

*Mathematics Subject Classification 1991:*
Primary 90C27; Secondary 68Q15

## 1. INTRODUCTION

The Traveling Salesman problem (TSP) is a well known problem. The salesman must travel to a number of cities, with each city visited exactly once before returning back to his home town. Such a tour is called a Hamiltonian tour. In the classical case of the Sum TSP (STSP) the objective is to minimize the total traveled distance, i.e.,



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to solve

$$[\text{STSP}] \quad \min \{\textstyle\sum_{i=1}^{n} d[i, \varphi(i)] \mid \varphi \text{ is a } \textit{Hamiltonian tour}\},$$

where $\varphi$ is a cyclic permutation of $\{1, 2, \ldots, n\}$ i.e., $\varphi(i)$ denotes the city next to city $i$ in the tour and $d[i, \varphi(i)]$ the distance from $i$ to $\varphi(i)$.

In case of the Bottleneck TSP (BTSP) the problem is to find the Hamiltonian tour that minimizes the longest traveled distance between two successive cities, i.e., to solve

$$[\text{BTSP}] \quad \min \{\max \{d[i, \varphi(i)] \mid i = 1, \ldots, n\} \mid \varphi \text{ is a } \textit{Hamiltonian tour}\}.$$

In the area of telecommunication research has been carried out about reliability of networks. In general reliability increases when instead of one, several cycles are constructed, where none of the cycles have an edge in common, i.e., the cycles exhibit edge-disjointness. This problem, where $K$ edge-disjoint Hamiltonian cycles of minimal total length are required, was introduced by Krarup [1975]; it is known as the $K$-Peripatetic Salesman Problem or K-PSP for short. De Kort [1993] has given a branch and bound algorithm for the 2-PSP.

Since TSP is a special case of the K-PSP (TSP is obviously the same as 1-PSP) the K-PSP is also in the class of NP-hard problems. Nevertheless there are instances of the TSP, some for the STSP, some for the BTSP and some for both, that are solvable in polynomial time, see, e.g., Gilmore *et al.* [1985]. We will investigate whether these cases remain well-solved for the corresponding 2-PSP, i.e., either the sum version (S2-PSP) or the bottleneck version (B2-PSP). We will show that the required edge-disjointness can be an obstacle when trying to solve the 2-PSP in polynomial time.

In graphs that are not complete, for instance a 4-regular graph (i.e., a graph with each node having a degree 4) even the existence of a PSP solution can already be non-trival. In a graph theory oriented context Barth and Raspaud [1994] have considered the butterfly graph, a 4-regular graph that can be seen as a model of a parallel computer architecture with the vertices representing processors and the edges communication lines between processors. They have proven that a 2-PSP solution exists in such a sparse graph.

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An alternative view on edge-disjoint tours is to see them as parallel tours, i.e., each salesman has to visit each city. In Multisalesmen or $k$-person problems each city has to be visited by one of the salesmen, i.e., solutions can be seen as a series of subtours. Pearn [1994] has shown solvable cases in the context of the $k$-person version of the Chinese Postman Problem. One of these cases is based on the notion of a *symmetric cycle:* a cycle is symmetric if the length in one direction equals the length in the other direction. For networks with only such cycles the TSP and the 2-PSP are equivalent.

A last introductory remark concerns a distinction in approach to be made between symmetric and asymmetric matrices. When dealing with symmetric matrices it will not be allowed to include an arc $(i, j)$ in one tour if the reversed arc $(j, i)$ is present in the other tour. Allowing this would lead to the trivial result that given an optimal solution to the TSP on some symmetric matrix, reversing this tour would yield a second, edge-disjoint tour of equal length and together these tours would form an optimal solution to the 2-PSP. One can alternatively consider a symmetric TSP as a TSP on a network with only symmetric Hamiltonian cycles.

In asymmetric matrices an arc $(i,j)$ in one tour and the reversed arc $(j,i)$ can be present in the other tour and the trivialities of the symmetric case will not arise; the reversed tour of an optimal solution of the TSP on an asymmetric matrix is not optimal in general.

## 2. SOME TRIVIAL RESULTS

In several well-solved special cases described by Gilmore *et al.* [1985] for the TSP it is obvious whether a solution of the 2-PSP can be determined in polynomial time. Because of their simplicity these cases will be dealt with briefly.

The first case to be mentioned is that of the *reducible networks*; it can be described as follows: cosider a directed graph $G=(V,A)$. A bipartition $(S,T)$ defines an *almost directed cut* if there is exactly one arc, say $(u,v)$, extending from a vertex $u$ in $T$ to a vertex $v$ in $S$. Obviously, when constructing a Hamiltonian cycle this specific arc must always be included and consequently some arcs of the network (namely all other arcs extending from $u$ and all other arcs ending in $v$)

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no longer have to be considered. So the network can be reduced to a simpler one by deleting these arcs, which in turn may create additional almost directed cuts.

A directed network is called *reducible* if it can be transformed to a tour by deleting arcs through the repeated discovery of almost directed cuts. Clearly a reducible network cannot contain two edge-disjoint Hamiltonian cycles since the one arc that characterizes a reducible network is contained in every Hamiltonian cycle of the network. Therefore a solution to the 2-PSP does not exist for reducible networks.

A second case is the existence of cutsets. An *edge cutset* of a connected graph is a minimal set of edges whose deletion leaves a disconnected graph. If it contains exactly $k$ edges, then it is a *k-edge cutset*. In case a graph contains a 3-edge cutset, it is impossible to construct two edge-disjoint Hamiltonian cycles, because a Hamiltonian cycle contains exactly two edges of any 3-edge cutset. So there is always one edge from any 3-edge cutset that is part of both Hamiltonian cycles. The conclusion is that a necessary though not sufficient condition for the existence of a solution to the 2-PSP in a certain graph is, that the graph contains a $k$-edge cutset ($k \geqslant 4$).

The last case in this section is that of the *constant* TSP. In this case two values are attached to each city: one for entering a city, $\mathbf{v}$, and one for leaving a city, $\mathbf{u}$. Thus the cost of traveling from city $i$ to city $j$ is given by: $c_{ij} = u_i + v_j$. Since every city always contributes the same cost to regardless which tour, namely the total value of entering and leaving that city, every existing tour has the same total cost. Therefore the myopic approach (solving the TSP, removing the edges used and solving the TSP again in the remaining graph) results in an optimal solution for the S2-PSP.

Although the length of a tour is constant, nothing can be said about the length of the individual edges; these are determined by the order in which the cities are visited. For example, if you were to go from the city with the highest outgoing value (max $\{u_i\}$) to the city with the highest incoming value (max $\{v_j\}$), you would have a tour with a very large edge and thus an inferior solution to the bottleneck problem. We think the B2-PSP to be NP-hard.

Without considering it further, we mention a mixed bottleneck sum criterion that minimizes the length of the longest tour of a K-PSP

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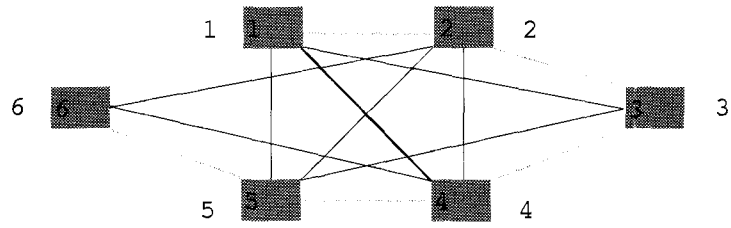



FIGURE 1 The myopic approach to 2-PSP.

problem. The constant PSP is also a trivial solvable case for this criterion as all the tours have the same length.

In general the myopic approach does not yield an optimal solution for the 2-PSP, as elucidated by Figure 1 taken from De Kort [1992]: the bold edge has length 3, the dotted edges have length 1 and all the other ones length 2. Solving the TSP yields the tour along dotted edges (1, 2, 3, 4, 5, 6, 1) with length 6. The only cycle that can be made of edges not in the first cycle is (1, 4, 6, 2, 5, 3, 1), with length 13, so the total length of the myopic 2-PSP solution is 19. An optimal 2-PSP solution is given by the cycles (1, 2, 6, 5, 4, 3, 1) and (1, 5, 3, 2, 4, 6, 1) with a total length of 18 ($<19$).

## 3. SMALL MATRICES

A matrix $C=((c_{ij}))$ is called *small* if there exist $n$-dimensional vectors $a$ and $b$ such that $c_{ij}=\min\{a_i, b_j\}$. A small matrix where all the $2n$ values of the vectors are distinct is said to have distinct values; such matrices are asymmetric. Let $d_i$ be the $i$th smallest of the $2n$ distinct values $a_i$ and $b_j$ and then let $D=\{d_1, d_2, \ldots, d_n\}$ with $d=\sum_{i=1}^{n} d_i$. The length of an optimal tour for both the STSP and the BTSP can be found easily and is limited to very few values, according to a result of Gabovich [1970]:

*Let C be a small matrix with distinct values. The length of an optimal* TSP tour for C is d if and only if one of the following three conditions holds:

(S1) *For at least one city i, both* $a_i$ *and* $b_i$ *are in D;*
(S2) $D=\{a_1, a_2, \ldots, a_n\}$;
(S3) $D=\{b_1, b_2, \ldots, b_n\}$.

We outline how to construct a tour.

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If (S2) or (S3) hold this is a constant TSP and can be treated that way (see section 2). Suppose (S1) holds; then the set of cities can be partitioned into four sets:

$\mathbf{D}_0$: the cities $i$ with neither $a_i$ nor $b_i$ in $D$;
$\mathbf{D}_2$: the cities $i$ with both $a_i$ and $b_i$ in $D$;
$\mathbf{D}_a$: the cities $i$ with only $a_i$ in $D$;
$\mathbf{D}_b$: the cities $i$ with only $b_i$ in $D$.

By a simple counting argument, $|D_0| = |D_2|$.

Construct now a tour as follows. Start with a city in $D_2$ (say $d_{21}$ in Fig. 2); then in any order visit the cities of $D_a$, followed by visiting the cities of $D_b$ in any order. The tour is completed by visiting the remaining cities in $D_2$ and $D_0$ alternately; that is, one from $D_2$, then one from $D_0$ and so on. It is easy to see that the cost of this tour is $d$. Since the values are distinct, each arc of the tour must have a different value, and thus the cost of any tour must be at least $d$. For more details we refer to, e.g., Gilmore [1985].

The 2-PSP on small matrices with distinct values can be solved in a similar way for both the sum and the bottleneck criterion. First we have to establish which of the three conditions holds (the case when none of the conditions hold will be treated further below). If either condition (S2) or (S3) holds we are dealing with the constant TSP (already treated in section 2). If condition (S1) holds, then to solve the TSP we can visit the cities in $D_a$ and $D_b$, in any order. The cities not yet in the tour after visiting all the cities in $D_a$ and $D_b$, can also be visited in any order as long as the cities in $D_2$ and $D_0$ are visited alternately.

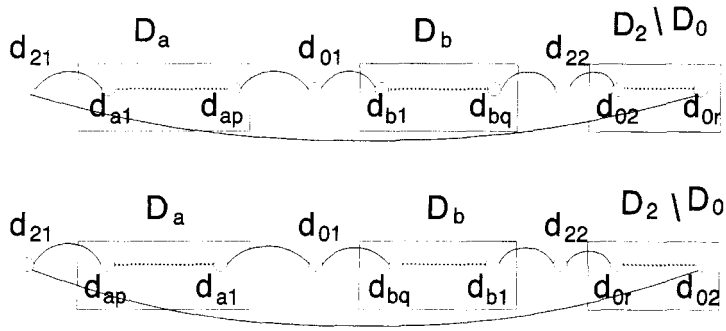



FIGURE 2 Two edge-disjoint tours that form an optimal solution to the 2-PSP ($D_a$, $D_b$ and $D_0$ have $p$, $q$ and $r$ elements, respectively).

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By constructing two edge-disjoint paths in both $D_a$ and $D_b$, and by constructing two such paths throught the remaining cities, visiting cities of $D_2$ and $D_0$ alternately, we obtain two edge-disjoint Hamiltonian cycles, both of length $d$, and thus an optimal solution to the S2-PSP. This construction is easy since we only need to reverse the path in subsets $D_a$ and $D_b$ and reverse the path starting from the first city of $D_0$ visited after all of the cities of $D_b$ have been visited, to obtain a second path that is edge-disjoint if $|D_a| \geqslant 2$ and $|D_b| \geqslant 2$.

This way of constructing a second tour also insures that the transition between different subsets is performed without violating the edge-disjointness restriction. This solution is obviously also bottleneck optimal since the two constructed cycles both contain the $n$ smallest values of the matrix, and the $n$th smallest edge/arc is a lower bound for the bottleneck problem. The solution is depicted in Figure 2; note that the path in the last box, $D_2 \backslash D_0$, visits alternately a city in $D_0$ and one in $D_2$.

For the case when a group of cities, for instance $D_a$, consists of only two cities, say $\{1, 2\}$, it is sufficient to note that small matrices are asymmetric and that the arc (1, 2) is disjoint with the arc (2,1) as a matter of definition.

If the conditions $S_1, S_2$ or $S_3$ fail to hold and therefore $d$ cannot be attained as the optimal value, then the optimal value follows from a further result of Gabovich [1970].

*Let $C$ be a small matrix with distinct values. The length of the shortest tour $z^*$ for the STSP on $C$ is either $d$ or $d - d_n + d_{n-1}$ or min $\{d - d_n + d_{n+2}, d - d_{n-1} + d_{n-1}\}$.*

*Furthermore, suppose that $z^* \neq d$; then $z^* > d - d_n + d_{n-1}$ if and only if one of the three following conditions holds (with $i'$ the largest index $i$ such that $a_i$ in $D$, the set of $n$ smallest values):*

(S4) $i' = 1$; $d_n = b_2$; $d_{n+1} = a_2$;
(S5) $i' = n - 1$; $d_n = a_{n-1}$; $d_{n+1} = b_{n-1}$;
(S6) $2 \leqslant i' \leqslant n - 2$; *either* ($d_n = a_{i'}$ and $d_{n+1} = b_{i'}$) *or* ($d_n = b_{i'+1}$ *and* $d_{n-1} = a_{i'+1}$).

Thus by checking conditions (S1), (S2),..., (S6) we can establish the value of $z^*$:

$z^* = d$ if either condition (S1), (S2) or (S3) holds;
$z^* = d - d_n + d_{n-1}$ if none of the conditions hold;
$z^* = \min \{d - d_n + d_{n-2}, d - d_{n-1} + d_{n+1}\}$ otherwise.

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To construct the solution to the S2-PSP we first replace the city associated with the $n$th or $(n-1)$th place by the city associated with the $(n+1)$th or $(n+2)$th place, depending on the value of $z^*$. With this new set of $n$ cities, divided into four subsets as before, we can construct two edge-disjoint Hamiltonian cycles the same way we did earlier in the case $z^* = d$.

For an optimal solution to the B2-PSP we have to be aware of the fact that if either condition (S4), (S5) or (S6) hold, we will always construct a tour where the city associated with value $d_{n+1}$ will replace the city associated with value $d_{n-1}$, because by definition $d_{n+1} \leqslant d_{n-2}$ and this replacement will therefore lead to a better bottleneck solution.

We have proven the following theorem:

THEOREM 1 *The S2-PSP as well as the B2-PSP are polynomially solvable on a small matrix with distinct values.*

## 4. CIRCULANT MATRICES

Even though a polynomial algorithm is not available for the STSP or the S2-PSP for circulant matrices, it can be shown under which circumstances a fast solution exists for asymmetric circulant matrices. A *circulant matrix* is a $(n \times n)$ matrix with the property that the entries $(i, j)$ with $(j-i) = k \pmod{n}$ have the same value $c_k$; these cells comprise the $k$th *stripe* of the matrix (of the form as given in Fig. 3); for symmetric matrices $c_k = c_{n-k}$. Each stripe yields a feasible solution to the assignment problem defined by the matrix, becaue every row as well as every column contains exactly one element of each stripe. Garfinkel [1977] showed the following result and the corollary:

*The number of subtours in the assignment given by the kth stripe is equal to the greatest common divisor of k and n*: gcd $(k, n)$.

COROLLARY *If n is prime then each stripe yields a Hamiltonian cycle except for the 0th (because gcd (0, n) is n by definition, that will never yield a Hamiltonian cycle).*

THEOREM 2 *The K-PSP on a graph with a prime number of cities is polynomially solvable on an asymmetric circulant edge weight matrix for both the sum and the bottleneck crietrion.*

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$$\begin{bmatrix} c_0 & c_1 & c_2 & \cdots & c_{n-2} & c_{n-1} \\ c_{n-1} & c_0 & c_1 & \cdots & c_{n-3} & c_{n-2} \\ c_{n-2} & c_{n-1} & c_0 & \cdots & c_{n-4} & c_{n-3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ c_1 & c_2 & c_3 & \cdots & c_{n-1} & c_0 \end{bmatrix} \qquad \begin{bmatrix} c_5 & c_4 & c_2 & c_3 & c_1 & c_3 & c_2 & c_4 \\ c_4 & c_5 & c_4 & c_2 & c_3 & c_1 & c_3 & c_2 \\ c_2 & c_4 & c_5 & c_4 & c_2 & c_3 & c_1 & c_3 \\ c_3 & c_2 & c_4 & c_5 & c_4 & c_2 & c_3 & c_1 \\ c_1 & c_3 & c_2 & c_4 & c_5 & c_4 & c_2 & c_3 \\ c_3 & c_1 & c_3 & c_2 & c_4 & c_5 & c_4 & c_2 \\ c_2 & c_3 & c_1 & c_3 & c_2 & c_4 & c_5 & c_4 \\ c_4 & c_2 & c_3 & c_1 & c_3 & c_2 & c_4 & c_5 \end{bmatrix}$$

FIGURE 3 Circulant matrices: (left) asymmetric, (right) symmetric, see example 1.

*Proof* Given $n$ is prime, an optimal solution to the K-PSP can be constructed as follows: order the stripes such that the value of stripe $k(i)$ is at least as large as the value of stripe $k(i-1)$: $c_{k(0)} \leqslant c_{k(1)} \leqslant \ldots \leqslant c_{k(n-1)}$. The smallest $K$ stripes form an optimal solution to the SK-PSP, because a lower bound for the K-PSP for the sum case is given by the sum of the Kn smallest edges in the graph. The stripes $k(0)$ up to $k(K-1)$ contain exactly these Kn smallest edges in the graph described by the Circulant Matrix.

A lower bound for the BK-PSP is the value of the Kn-th smallest edge in the weight matrix. The Kn-th smallest edge is the largest edge in the obtained solution; it is an edge from stripe $k(K-1)$. Therefore the stripes form an optimal solution to the BK-PSP as well. □

In the symmetric case this method would result in a solution into which both the smallest stripe and its symmetric counterpart are included, since they have the same costs. But this is prohibited, as in the symmetric case a tour and its reverse are not considered to be edge disjoint.

The algorithm of Burkard and Sandholzer [1991] for solving the BTSP on symmetric circulant matrices works as follows: order the stripes in order of non-decreasing costs and let

$$t^* = \min\{t: g_t = \gcd(n, k(0), k(1), \ldots, K(t)) = 1\};$$

an optimal solution to the BTSP can be constructed by patching of these stripes. For the B2-PSP however this approach fails as is elucidated by example 1:

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*Example 1* Consider the symmetric circulant matrix in Figure 3, with $c_1 < c_2 < \ldots < c_5$ and stripes called $k(1)$ - with the lowest costs -, $k(2)$ - with the second lowest costs - and so on.

According to the method of Burkard & Sandholzer [1991], the stripe with the lowest costs, $k(1)$, is patched up by using more expensive edges from $k(2)$. Then the subcycles $(1,5,1)$ and $(3,7,3)$ are patched up to form the cycle $(1,3,7,5,1)$, while the subcycles $(2,6,2)$ and $(4,8,4)$ are transformed into the cycle $(2,4,8,6,2)$. Proceeding in this manner we find an optimal Hamiltonian cycle for the BTSP: $(1,4,8,6,2,5,7,3,1)$.

The construction of this tour uses edges of the stripes $k(1)$, $k(2)$ and $k(3)$. The edges of these stripes not yet used are $(3,5)$, $(1,7)$, $(4,6)$ and $(2,8)$, the edges from the paths $(4,7,2)$ and $(5,8,3,6,1)$ and all their symmetric counterparts. Unfortunately, no second tour can be constructed with these edges, because the only way to include city 1 as well as city 4 in a tour is in between cities 6 and 7. Therefore constructing a tour is impossible.

Two edge-disjoint Hamiltonian cycles containing only the edges from stripes $k(1)$, $k(2)$ and $k(3)$ do exist, namely the cycles $(1,3,8,6,4,2,5,7,1)$ and $(1,4,7,2,8,5,3,6,1)$. This goes to show that the method of Burkard & Sandholzer [1991] is unable to solve the B2-PSP.

So it is open whether the B2-PSP is NP-hard on a symmetric circulant weight matrix.

## 5. ORDERED PRODUCT MATRICES

A matrix $D$ is called a *product matrix* if $D = A \cdot B^t$ with $A = (a_i)$ and $B = (b_i)$ two $n$-dimensional real vectors. We will assume the ordering: $a_1 \leqslant a_2 \leqslant \ldots \leqslant a_n \wedge b_1 \geqslant b_2 \geqslant \ldots \geqslant b_n$.

The STSP on such a matrix, called an *ordered product matrix* (OPM), can be solved in polynomial time; generalizing this algorithm for the S2-PSP requires, just as in the case of Graded Matrices, solving two assignment problems simultaneously, so it remains open whether a polynomial algorithm exists in this case. However for the B2-PSP on an OPM there are cases that can be solved in polynomial time.

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Van der Veen [1993] has shown that an OPM is or can be transformed into a *doubly graded matrix* or a *max-distribution matrix* of a special type. We consider related solvable cases for the 2-PSP.

### 5a. Diagonally Graded Matrices

A matrix $C = ((c_{ij}))$ is *graded across its rows* (*up its columns*) if

$$c_{ij} \leqslant c_{i,j+1} \, (c_{ij} \geqslant c_{i+1,j}) \, \forall \, i, j.$$

A matrix is called *doubly graded* if the matrix or a transpose of it is both graded across its rows and up its columns.

Now we introduce a related notion:
a matrix is *k-diagonally graded* if the matrix is doubly graded and $c_{i,i+k} \geqslant c_{i+1,i+k+1} \, \forall \, i < n-k$.

The SK-PSP on a $K$-diagonally graded matrix is NP-hard, because any matrix can by transformed into a $K$-diagonally graded one by adding suitably large numbers to rows and columns; such a transformation does not change the optimal solution set for the sum criterion, while it changes in general this set for the bottleneck criterion.

THEOREM 3 *The B2-PSP on a 2-diagonally graded matrix can be solved polynomially.*

*Proof* The BTSP on doubly graded matrices is solved by the tour $(1, 2, \ldots, n)$, see Gilmore [1985]; so the same holds for the BTSP on a 2-diagonally graded matrix.

To construct a second Hamiltonian cycle that is edge-disjoint with the first one we have to use a second entry from the first row. To make the solution bottleneck optimal we only have to select the smallest entry not yet used. This entry is easy to find: it is by definition of 2-diagonally graded matrices the arc $(1, 3)$. By the same definition the other entries of the diagonal on which arc $(1, 3)$ lies are smaller, so they all, together with the entries on the diagonals to the left, who are smaller as well, can be used to construct a second Hamiltonian cycle. These entries will yield the Hamiltonian cycle (whose largest entry is arc $(1, 3)$):

$$(1, 3, 5, \ldots, n, 2, 4, \ldots, n-1) \qquad \text{if } n \text{ is odd,}$$

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$$\begin{pmatrix} 9 & 11 & 14 & 17 & 19 \\ 7 & 9 & 12 & 13 & 18 \\ 6 & 6 & 7 & 10 & 12 \\ 3 & 4 & 5 & 8 & 11 \\ 1 & 2 & 2 & 6 & 9 \end{pmatrix}$$

FIGURE 4 Example of a 2-diagonally graded matrix.

$$(1, 3, 5, \ldots, n-3, 2, 4, \ldots, n, n-1) \qquad \text{otherwise.}$$

Together with the cycle $(1, 2, \ldots, n)$ this yields an optimal solution to the B2-PSP with bottleneck value $c_{13}$. □

*Example 2* Consider the matrix in Figure 4; an optimal solution to the B2-PSP is given by the tours: $t^1 = (1, 2, 3, 4, 5)$ and $t^2 = (1, 3, 5, 2, 4)$ with the bottleneck value $c_{13} = 14$.

### 5b. Max-Distribution Matrices

A matrix $D$ is called a *max-distribution matrix* if $\max\{d_{ij}, d_{kl}\} \leqslant \max\{d_{il}, d_{kj}\}$ for all $i < k$ and all $j < l$.

A tour is *pyramidal* if it is of the form

$$(1 = i_1, \ldots, i_r, n = j_1, \ldots, j_{n-r})$$

where $i_1 < i_2 < \ldots < i_r$ and $n = j_1 > j_2 \ldots > j_{n-r}$. Such a tour is essential in the following result found in Burkard and Sandholzer [1991]:

*For a BTSP on a max-distribution matrix a pyramidal tour exists that is optimal.*

Now we introduce some definitions and notations for a max-distribution matrix, say $D$:

$$\underline{A} := \{i | a_i < 0\} \text{ and } \underline{A}^c := \{i | a_i \geqslant 0\}, \quad \underline{B} := \{i | b_i \geqslant 0\}$$

with $\alpha := |\underline{A}|$ and $\beta := |\underline{B}|$, i.e., $\alpha$ is the number of negative $a_i$ values and $\beta$ the number of nonnegative $b_i$ values. Furthermore, define:

$$n_1 := \min\{\alpha, \beta\}, n_2 := |\alpha - \beta|, n_3 := n - \max\{\alpha, \beta\}, \text{ and } m := n_1 + n_2.$$

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The matrix $D$ can be partitioned with $D_{ij}$ a $(n_i \times n_j)$—matrix for $i, j = 1, 2, 3$:

$$D = \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ D_{21} & D_{22} & D_{23} \\ D_{31} & D_{32} & D_{33} \end{bmatrix}.$$

Some submatrices may vanish, e.g., if $A > 0$, $n_1 = 0$ and hence only $D_{22}, D_{23}, D_{32}$ and $D_{33}$ exist.

When trying to solve the B2-PSP on max-distribution matrices we distinguish between the cases $\alpha = \beta$ and $\alpha < \beta$. Van der Veen [1993] has shown that the case $\alpha > \beta$ can be treated in the same way as the case $\alpha < \beta$.

*Case* $\alpha = \beta$.

Since $n_2 = 0$, the partitioning of $D$ becomes

$$D = \begin{bmatrix} D_{11} & D_{13} \\ D_{31} & D_{33} \end{bmatrix}$$

with $D_{11} \leqslant 0, D_{13} \geqslant 0, D_{31} \geqslant 0$ and $D_{33} \leqslant 0$; furthermore, $D_{13}$ and $D_{31}$ are doubly graded.

Because $D_{11}$ and $D_{33}$ only have non-positive entries, the length of a Hamiltonian tour is determined by the arcs from a vertex in $\underline{A}$ to a vertex in $\underline{A}^c$ and vice versa. As $D$ is a max-distribution matrix, we only have to consider pyramidal tours according to the result on the BTSP. Moreover, we can restrict ourselves to pyramidal tours using exactly one arc from a vertex in $\underline{A}$ to a vertex in $\underline{A}^c$ and exactly one arc in the other direction. On top of this, since $D_{13}$ and $D_{31}$ are doubly graded only the following cases have to be considered ($\mathrm{bd}(t^i)$ is the bottleneck value of tour $t^i$):

1. $t^1 = (1 \text{ to } n_1 - 1, n_1 + 1, n \text{ to } n_1 + 2, n_1)$ with

$$\mathrm{bd}(t^1) = \max\{d[n_1 - 1, n_1 + 1], d[n_1 + 2, n_1]\};$$

2. $t^2 = (1 \text{ to } n_1 - 1, n_1 + 2 \text{ to } n, n_1 + 1, n_1)$ with

$$\mathrm{bd}(t^2) = \max\{d[n_1 - 1, n_1 + 2], d[n_1 + 1, n_1]\};$$

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3. $t^3 = (1, n_1, n_1 + 2 \text{ to } n, n_1 + 1, n_1 - 1 \text{ to } 2)$ with

$$\text{bd}(t^3) = \max\{d[n_1, n_1 + 2], d[n_1 + 1, n_1 - 1]\};$$

4. $t^4 = (1, n_1, n_1 + 1, n \text{ to } n_1 + 2, n_1 - 1 \text{ to } 2)$ with

$$\text{bd}(t^4) = \max\{d[n_1, n_1 + 1], d[n_1 + 2, n_1 - 1]\};$$

The four tours are depicted in Figure 5; the boxes represent the sets $\underline{A}$ and $\underline{A}^c$, while wiggly edges between vertices $i$ and $j$ represent the paths $[i \text{ to } j]$.

If $n_1 = 1$ or $n_1 = n - 1$ then the number of cases can be further reduced to two. Note that $t^1$ is the reverse tour of $t^3$ and $t^4$ the reverse tour of $t^2$.

The tour with a smallest bottleneck value is optimal for the BTSP. But since that bottleneck edge is fundamental and unique to that tour,

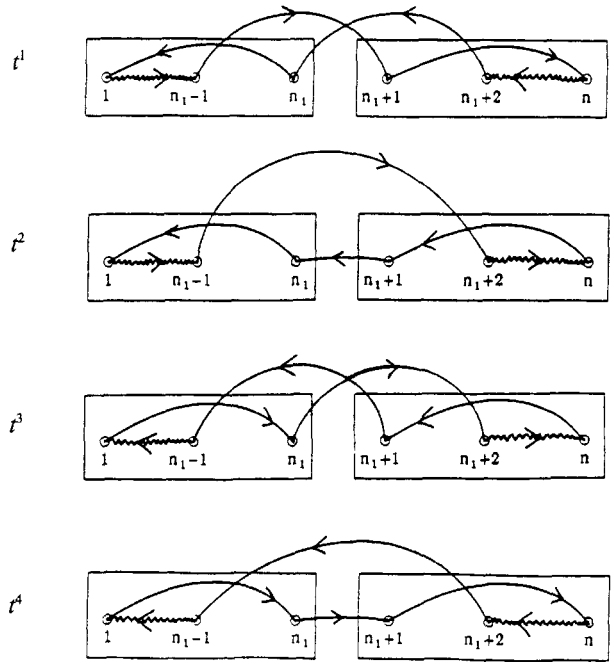



FIGURE 5 The tours $t^1$ up to $t^4$ (cf. Van der Veen [1993]).

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a solution to the B2-PSP cannot involve two tours with the same bottleneck-edge. Fortunately, the four listed tours are the four tours with the smallest bottleneck values. So the two tours with the smallest bottleneck value could yield the solution to the B2-PSP. Furthermore, these tours can be made edge-disjoint by the following permutations that will not affect the bottleneck value:

— if one tour is the reverse of the other, nothing has to be done, as they are already edge-disjoint;
— if the two tours have the path (1 to $n_1 - 1$) and the edge $(n_1, 1)$ in common, transform one of the tours to contain the new path ($n_1 - 2$ to $1, n_1 - 1$) and substitute the edge $(n_1, 1)$ by $(n_1, n_1 - 2)$;
— if the two tours have the path ($n$ to $n_1 + 2$) and the edge $(n_1 + 1, n)$ in common, transform one of the tours to contain the new path ($n_1 + 3$ to $n$) and substitute the edge $(n_1 + 1, n)$ by $(n_1 + 1, n_1 + 3)$.

No new bottlenecks can occur in the new tours because no edges from a vertex in $\underline{A}$ to a vertex in $\underline{A}^c$, or vice versa, have been inserted.

Thus, an optimal solution to the B-2PSP is constructed in two steps:

— select the two tours with the smallest bottleneck value;
— transform these tours into edge-disjoint ones, if necessary, as described.

We have proven the following theorem:

THEOREM 4 *The B2-PSP is polynomially solvable on a max-distribution matrix with the number of negative a-values equal to the number of non-negative b-values.*

The result is illustrated by an example: (Van der Venn [1993])

*Example 3* Let $A = [-4 \; -3 \; -2 \; 1 \; 5]$ and $B = [7 \; 3 \; 1 \; -2 \; -3]$; $\alpha = \beta = 3$;

$$D = \begin{bmatrix} -28 & -12 & -4 & 8 & 12 \\ -21 & -9 & -3 & 6 & 9 \\ -14 & -6 & -2 & 4 & 6 \\ 7 & 3 & 1 & -2 & -3 \\ 35 & 15 & 5 & -10 & -5 \end{bmatrix};$$

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$$\text{bd}(t^1) = \max\{d[2,4], d[5,3]\} = 6; \text{bd}(t^2) = \max\{d[2,5], d[4,3]\} = 9;$$

$$\text{bd}(t^3) = \max\{d[4,2], d[3,5]\} = 6; \text{bd}(t^4) = \max\{d[5,2], d[3,4]\} = 15.$$

So tours $t^1$ and $t^3$ have the smallest bottleneck values, therefore

$$t^1 = (1,2,4,5,3) \text{ and } t^3 = (1,3,5,4,2)$$

form an optimal solution to the B2-PSP with critical value 6.

*Case* $\alpha < \beta$.

In this case $\min\{n_1, n_2, n_3\} > 0$, so all the submatrices in the partitioning of matrix $D$ exist. Furthermore: $D_{11}, D_{12}, \leqslant 0$, $D_{13}, D_{21}, D_{22} \geqslant 0$, $D_{23} \leqslant 0$, $D_{31}, D_{32} \geqslant 0$, $D_{33} \leqslant 0$.

Decompose the vertex set into three disjoint sets

$$V_1 := \{1, \ldots, n_1\}, V_2 := \{n_1 + 1, \ldots, m\} \text{ and } V_3 := \{m+1, \ldots, n\}.$$

From the structure of $D$ as described, we have the following. There are no positive costs to travel within the sets $V_1$ and $V_3$, no positive costs to go from $V_1$ to $V_2$, and to go from $V_2$ to $V_3$, and within $V_2$ there are no positive costs to go from a vertex with a lower index to a vertex with a higher index. So the length of a Hamiltonian tour is determined by the arcs from a vertex in $V_1$ to a vertex in $V_3$ or vice versa, arcs from a vertex in $V_3$ to a vertex in $V_2$, arcs from a vertex in $V_2$ to a vertex in $V_1$ and the arcs in $V_2$ from a vertex with a higher index to a vertex with a lower index.

Van der Veen [1993] proved as lemma:

*A shortest path from n to 1 is the Hamiltonian path [n to 1], with length*

$$L := max\{d[i, i-1] \mid i = n_1 + 1, \ldots, m+1\}.$$

Since $L$ is a lower bound for the BTSP, since we only have to consider pyramidal tours ($D$ is a max-distribution matrix), since $D_{22}$ is lower triangular, and since the nonnegative submatrices are doubly graded, it follows that we only have to consider five cases:

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(1) Let $v$ be the vertex in $\{n_1+2,\ldots,m+1\}$ that minimizes $d[i,i-2]$;
$t^1=(1,v-1,n \text{ to } v,v-2 \text{ to } 2)$
with $\mathrm{bd}(t^1)=\max\{d[v,v-2],L\}$;

(2) $t^2=(1 \text{ to } n_1-1,m+2 \text{ to } n,m+1 \text{ to } n_1)$
with $\mathrm{bd}(t^2)=\max\{d[n_1-1,m+2],L\}$;

(3) $t^3=(1 \text{ to } n_1-1,m+1,n \text{ to } m+2,m \text{ to } n_1)$
with $\mathrm{bd}(t^3)=\max\{d[m+2,m],d[n_1-1,m+1],L\}$;

(4) $t^4=(1,n_1,m+2 \text{ to } n,m+1 \text{ to } n_1+1,n_1-1 \text{ to } 2)$
with $\mathrm{bd}(t^4)=\max\{d[n_1+1,n_1-1],d[n_1,m+2],L\}$;

(5) $t^5=(1,n_1,m+1,n \text{ to } m+2,m \text{ to } n_1,n_1-1 \text{ to } 2)$
with $\mathrm{bd}(t^5)=\max\{d[n_1+1,n_1-1],d[m,m+2],\ d[n_1,m+1],L\}$.

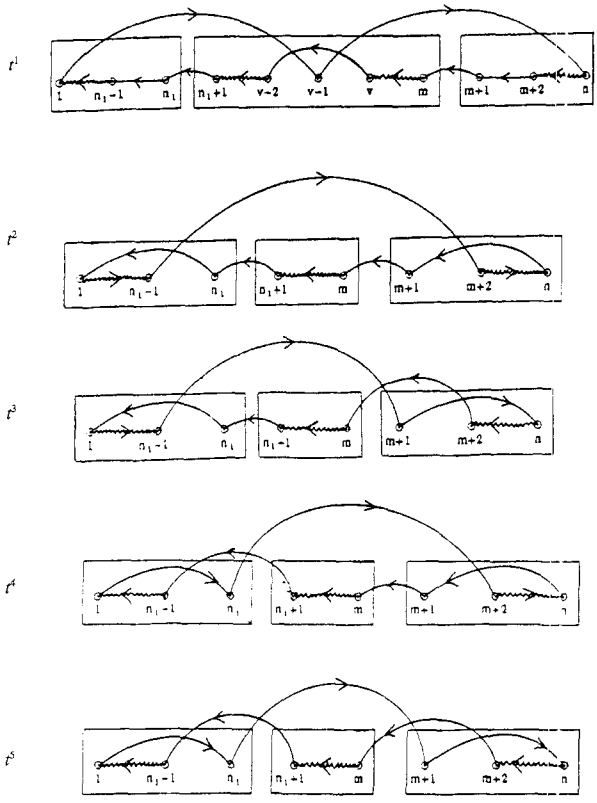



FIGURE 6 The tours $t^1$ up to $t^5$ (cf. Van der Veen [1993]).

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Computing the shortest of these five tours can be done in polynomial time. The tours are depicted in Figure 6; the boxes represent the sets $V_1, V_2$ and $V_3$.

Like in the case $\alpha = \beta$ we are searching for the shortest two tours to solve the B2-PSP. Unfortunately, in contrast to the case $\alpha = \beta$, the tours in this case cannot be made edge-disjoint. This is due to the fact that if we try to construct a second edge-disjoint path in $V_2$, the subset in which all five tours have, at least for a great part, the path ($m$ to $n_1$) in common (as can be seen in Fig. 6), we have to introduce a number of edges $(v, v-i)$ with $v \in \{m, \ldots, n_1 + 3\}$ and $i \in \{2, \ldots, m - n_1 - 1\}$. In fact, this would mean we are searching for a shortest edge-disjoint Hamiltonian path in $V_2$; such a search however, cannot be completed generally in polynomial time (see Garey and Johnson [1979]).

Thus it can be concluded that the case $\alpha < \beta$ (and similarly the case $\alpha > \beta$) does not offer the possibility of solving the B2-PSP in polynomial time.

## 6. REMARKS AND CONCLUSIONS

For a number of well-solved cases for the TSP, the first three as described in Gilmore *et al.* [1985], we have tried to generalize them for the 2-PSP, however without success; we mention:

— problems on upper triangular cost matrices;
— problems on graded cost matrices;
— problems on cost matrices obeying the Demidenko conditions; where the TSP has a pyramidal tour as optimal solution, this cannot be said for the 2-PSP;
— convex-hull-and-line problems (see Deineko, Van Dal and Rote [1994]); then one needs two edge disjoint paths, an NP-hard problem.

It is open whether these special cases are NP-hard for the PSP.

As a conclusion we can say that it is possible to polynomially solve the 2-PSP in the following non-trivial cases:

— problems on Small matrices with distinct values both for the sum as well as for the bottleneck criterion;

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— problems on asymmetric Circulant matrices with a prime number of cities again for both criteria;
— problems on 2-diagonally graded matrices for the bottleneck criterion;
— problems for the bottleneck criterion on max-distribution matrices with the number of negative $a$-values equal to the number of non-negative $b$-values.

In most well-solved cases it is open for further research whether the K-PSP for the value of K larger than 2 remains polynomially solvable.

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